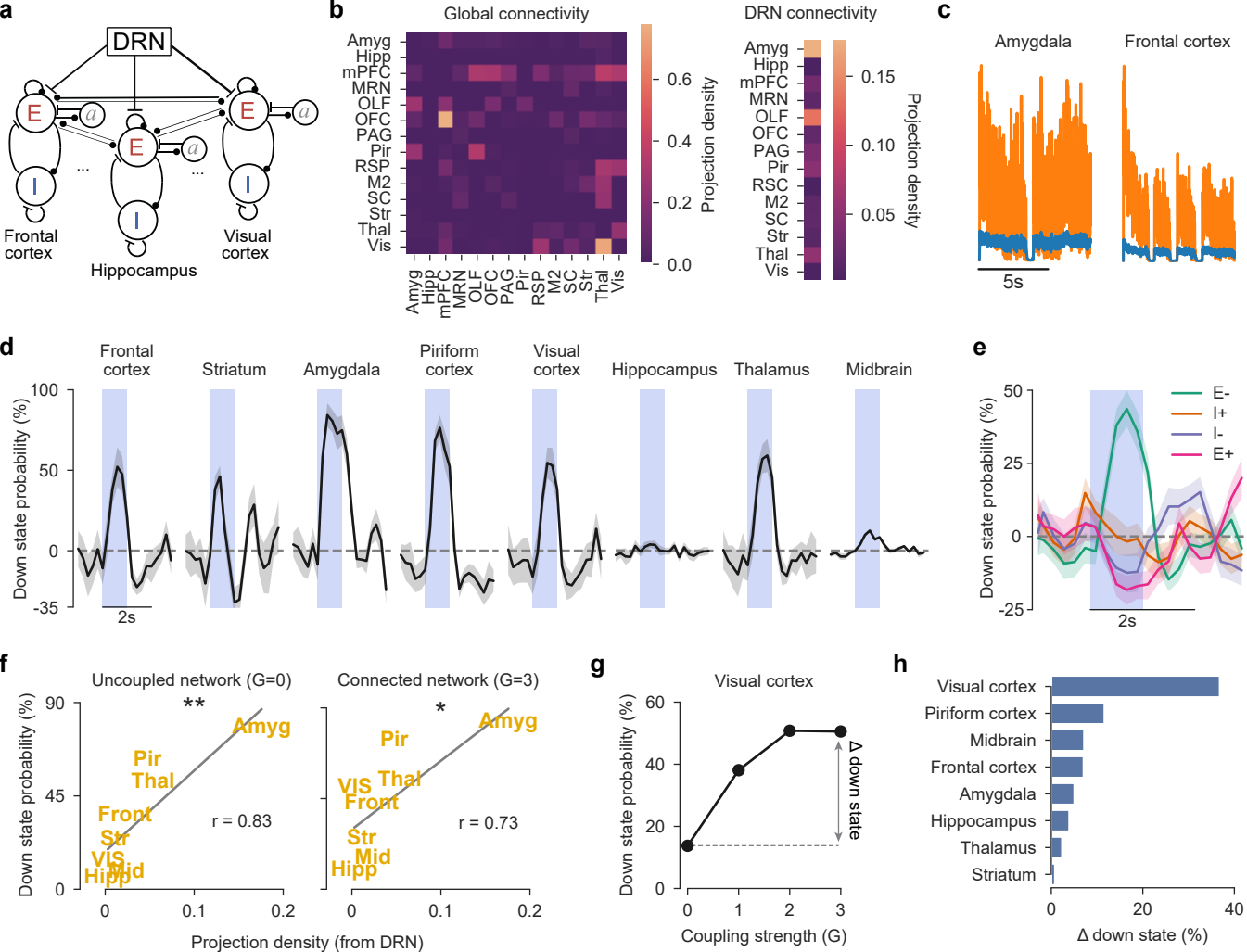

a
DRN
E
a
I
Frontal cortex
Visual cortex
Hippocampus
b
Global connectivity
Amyg
Hipp
mPFC
MRN
OLF
OFC
PAG
Pir
RSP
M2
SC
Str
Thal
Vis
Projection density
0.6
0.4
0.2
0.0
DRN connectivity
0.15
0.10
0.05
c
Amygdala
Frontal cortex
5s
d
Frontal cortex
Striatum
Amygdala
Piriform cortex
Visual cortex
Hippocampus
Thalamus
Midbrain
Down state probability (%)
100
50
0
-35
2s
e
E-
I+
I-
E+
50
25
0
-25
f
Uncoupled network (G=0)
Connected network (G=3)
**
*
r = 0.83
r = 0.73
Front
Mid
90
45
0.1
0.2
Projection density (from DRN)
g
Visual cortex
60
40
20
Δ down state
Coupling strength (G)
h
Δ down state (%)
40